

DECODELABS INTERNSHIP



PROJECT 2: STUDENT PERFORMANCE DATA ANALYSIS

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Objective

The objective of this project is to analyze student performance data using Microsoft Excel. The project involves data cleaning, organizing data, generating summaries, and creating visualizations to gain meaningful insights.

Software Used

- Microsoft Excel
- GitHub

Introduction

Data analysis is the process of examining, cleaning, transforming, and presenting data to discover useful information. In this project, student performance data was analyzed using Microsoft Excel. The dataset contained student information such as gender, marks, attendance, and performance records.

Dataset Description

The dataset contains:

- Student ID
- Gender
- Attendance Percentage
- Marks Obtained
- Performance Information

The data was examined for missing values and duplicate records before analysis.

Data Cleaning Process

The following data cleaning operations were performed:

1. Removed duplicate records.
2. Checked for missing values.
3. Corrected formatting issues.
4. Standardized column names.
5. Verified data consistency.

Data Analysis

The following analyses were performed:

1. Total Number of Students

Counted the total students present in the dataset.

2. Gender Distribution

Calculated the number of male and female students.

3. Average Marks

Computed average marks of all students.

4. Attendance Analysis

Analyzed attendance percentages and identified attendance trends.

5. Performance Analysis

Classified students based on academic performance.

Visualization

The following charts were created in Excel:

Pie Chart

Shows gender distribution among students.

Bar Chart

Shows comparison of student marks.

Attendance Analysis Chart

Displays attendance percentages.

Performance Summary Chart

Illustrates performance categories of students.

Results

The analysis provided the following insights:

- Total number of students was identified.
- Gender distribution was visualized.
- Average marks were calculated.
- Attendance trends were analyzed.
- Student performance patterns were identified.

Conclusion

This project successfully demonstrated the use of Microsoft Excel for data cleaning, analysis, and visualization. The generated charts and summaries helped in understanding student performance and attendance patterns. The project improved practical skills in Excel-based data analytics and reporting.

References

1. Microsoft Excel Documentation
2. Student Performance Dataset
3. DecodeLabs Internship Guidelines

